Thomas C. Fraser

Curriculum Vitae

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Personal

Email	tcf@math.ku.dk
Webpage	www.tcfraser.com
Phone	DEN: +45 81 94 61 22 CAN: +1 226 868 0557
Address	Strynøgade 5, 1.08, 2100 Copenhagen, Denmark
Citizenship	Canadian
Date of Birth	27/11/1994

Positions & education

Jan. 2024 – present	Postdoctoral Researcher Department of Mathematics, University of Copenhagen
Sept. 2018 – Sept. 2023	Ph.D., Physics Thesis: An estimation theoretic approach to quantum realizability problems Perimeter Institute For Theoretical Physics, Waterloo, ON
Aug. 2017 – Jun. 2018	M.Sc., Theoretical Physics, Quantum Information Thesis: A Combinatorial Solution to Causal Compatibility Perimeter Scholars International Program, Perimeter Institute For Theoretical Physics, Waterloo, ON
Sept. 2012 – Apr. 2017	B.Sc., Mathematical Physics, Astrophysics Specialization University of Waterloo, Waterloo, ON

Papers

2026	Complete entanglement detection using polynomial invariants Thomas C. Fraser, V. Blakaj, R. Rubboli, F. Huber, M. Fanizza arXiv:2606.16712
2026	On quantum functionals for higher-order tensors A. Botero, M. Christandl, Thomas C Fraser, I. Leigh, H. Nieuwboer arXiv:2604.18283
2025	The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols A. Alekseev, M. Christandl, Thomas C. Fraser arXiv:2510.04877
2023	An estimation theoretic approach to quantum realizability problems Thomas. C. Fraser Ph.D. Thesis. arXiv:2402.10902
2023	Understanding the interplay of entanglement and nonlocality: motivating and developing a new branch of entanglement theory D. Schmid, Thomas. C. Fraser, R. Kunjwal, A. B. Sainz, E. Wolfe, R. W. Spekkens Quantum 7, 1194

2022 A sufficient family of necessary inequalities for the compatibility of quantum marginals
Thomas. C. Fraser
arXiv preprint arXiv:2211.00685

2020 A Combinatorial Solution to Causal Compatibility
Thomas. C. Fraser
Journal of Causal Inference 8(1), 22-53

2018 Machine Learning Peeling and Loss Modelling of Time-Domain Reflectometry
J. R. Rinehart, J. H. Béjanin, Thomas. C. Fraser, M. Mariantoni
arXiv preprint arXiv:1804.04756

2018 Causal Compatibility Inequalities Admitting of Quantum Violations in the Triangle Scenario
Thomas. C. Fraser, E. Wolfe
Physical Review A, 98(2), 022113

Teaching

Block 3 2025-2026	Introduction to Representation Theory - University of Copenhagen Lecturer
Block 3 2024	Introduction to Representation Theory - University of Copenhagen Teaching Assistant
Winter 2021	Quantum Information - Perimeter Scholars International (PSI) Program Teaching Assistant
Winter 2021	Quantum-classical gaps in causal structures - PSI Winter School Student project co-lead
Fall 2018	Physics 1 - Phys 111 - University of Waterloo Teaching Assistant

Work experience

September 2015 – January 2016	Research & Development Data Scientist Meltwater (formerly Sysomos), Toronto, ON, Canada
September 2014 – May 2015	Video Game Developer Lunarch Studios, Waterloo, ON, Canada
January 2014 – May 2014	Mathematics Tutor Humber College, Toronto, ON, Canada
June 2011 – August 2011	Solar Panel Technician OVG Solar, Inc, Renfew, ON, Canada

Programming

Languages	C, C++, Python, Mathematica, Lua, Matlab, Actionscript, JavaScript